

Oxidation - Reduction

- What is the oxidizing agent in the following reaction?

$$\text{Fe}(\text{CN})_6^{3-} + \text{Co}(\text{CN})_5^{3-} + \text{CN}^- \rightarrow \text{Fe}(\text{CN})_6^{4-} + \text{Co}(\text{CN})_6^{3-}$$

(A) $\text{Fe}(\text{CN})_6^{3-}$ (B) $\text{Co}(\text{CN})_5^{3-}$
 (C) CN^- (D) $\text{Co}(\text{CN})_6^{3-}$
- What are the oxidation states of N in NH_4NO_3 ?
 (A) Both are -3
 (B) Both are $+1$
 (C) One is -1 and one is $+1$
 (D) One is -3 and one is $+5$
- A 0.4679 g sample of iron-containing ore is dissolved in acid and the iron reduced to the $+2$ oxidation state. The solution is titrated with 0.03930 M KMnO_4 , requiring 26.82 mL to reach the equivalence point. What is the percentage by mass of iron in the ore?
 (A) 12.58% (B) 29.43%
 (C) 62.91% (D) 72.36%
- Ships sometimes have strips of titanium welded to their steel hulls to protect them from corrosion. Which is the best explanation for this protective action?
 (A) Ti is more easily oxidized than Fe.
 (B) Ti and Fe form an alloy that stabilizes the Fe.
 (C) Ti^{2+} has a more positive reduction potential than Fe^{2+} .
 (D) The Ti and Fe form a galvanic cell with Ti being reduced.
- Hypophosphorous acid (H_3PO_2) disproportionates in acidic solution to give phosphine (PH_3) and phosphorous acid (H_3PO_3). What ratio of $\text{PH}_3 : \text{H}_3\text{PO}_3$ is produced?
 (A) 1 : 1 (B) 1 : 2 (C) 2 : 3 (D) 3 : 2
- A solution containing 0.0100 mol of silver nitrate and 0.0100 mol copper(II) nitrate is electrolyzed with a current of 0.170 A for 152 minutes. What mass of metal is deposited on the cathode?
 (A) 1.12 g (B) 1.27 g
 (C) 1.38 g (D) 1.73 g
- What is the average oxidation state of iron in halotrichite, $\text{FeAl}_2(\text{SO}_4)_4 \cdot 22 \text{H}_2\text{O}$?
 (A) 0 (B) $+2$ (C) $+2.5$ (D) $+3$
- Which half-reaction takes place at the anode during electrolysis of aqueous sodium iodide?
 (A) $\text{Na}^+(\text{aq}) + e^- \rightarrow \text{Na}(\text{s})$
 (B) $2 \text{H}_2\text{O}(\text{l}) + 2 e^- \rightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
 (C) $2 \text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{s}) + 2 e^-$
 (D) $2 \text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^-$
- In an electrochemical cell, the reaction shown below occurs. Which changes will decrease the cell potential?

$$\text{Fe}^{2+}(\text{aq}) + \text{Ag}^+(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$$

I. Decreasing the concentration of $\text{Fe}^{2+}(\text{aq})$
 II. Increasing the size of the silver electrode

(A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
- 180 mL of a solution that is 0.024 M in both AgNO_3 and $\text{Cu}(\text{NO}_3)_2$ is electrolyzed with a current of 0.70 A for 350 s. What deposits on the cathode?
 (A) Ag only
 (B) Cu only
 (C) 2.5×10^{-3} mol Ag and some Cu
 (D) 2.5×10^{-3} mol Cu and some Ag
- What is the standard potential of a fuel cell where methanol is oxidized to carbon dioxide and oxygen is reduced to water?

Half-reaction	E° , V
$\text{CO}_2(\text{g}) + 6 \text{H}^+(\text{aq}) + 6 e^- \rightarrow \text{CH}_3\text{OH}(\text{l}) + \text{H}_2\text{O}(\text{l})$	-0.02
$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^- \rightarrow 2 \text{H}_2\text{O}(\text{l})$	1.23

(A) 0.30 V (B) 0.31 V
 (C) 1.21 V (D) 1.25 V
- The half-cell potential at 298 K for the reduction of an oxoanion becomes more negative by 0.118 V for each increase of 1 pH unit. How many protons and electrons are involved in this half-reaction?
 (A) 1 proton, 1 electron (B) 1 proton, 2 electrons
 (C) 2 protons, 1 electron (D) 3 protons, 2 electrons
- What is the oxidation number of carbon in methanol, CH_3OH ?
 (A) -2 (B) 0 (C) $+2$ (D) $+4$

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14. A standard galvanic cell is assembled from silver and platinum in 1 M solutions of the corresponding metal ions. Which metal would be labeled the cathode, and what is the voltage of the standard galvanic cell?

Half-reaction	E° , V
$\text{Pt}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Pt}(\text{s})$	1.20
$\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag}(\text{s})$	0.80

- (A) Ag cathode, $E^\circ = 1.60$ V
 (B) Pt cathode, $E^\circ = 1.20$ V
 (C) Ag cathode, $E^\circ = 0.80$ V
 (D) Pt cathode, $E^\circ = 0.40$ V
15. A spoon is electroplated in a solution of 1.0 M silver nitrate with a current of 0.10 A. How long does it take to plate 0.10 g of silver on the spoon?
- (A) 5.0 min (B) 10. min
 (C) 15. min (D) 30. min
16. Aqueous sulfuric acid containing 1.0 M NaI is electrolyzed at glassy carbon electrodes. What is produced at the anode?
- (A) H_2 (B) O_2
 (C) I_2 (D) SO_2
17. A certain galvanic cell has a standard cell potential that increases with increasing temperature. Which best explains this observation?
- (A) $\Delta H^\circ_{\text{rxn}} > 0$ (B) $\Delta S^\circ_{\text{rxn}} > 0$
 (C) $\Delta G^\circ_{\text{rxn}} > 0$ (D) $E^\circ > 0$
18. What is the standard reduction potential under acidic conditions for the reduction of $\text{PbO}_2(\text{s})$ to $\text{Pb}(\text{s})$?
- $\text{PbO}_2(\text{s}) + 4 \text{H}^+(\text{aq}) + 4e^- \rightarrow \text{Pb}(\text{s}) + 2 \text{H}_2\text{O}(\text{l}) \quad E^\circ = ???$
- | Half-reaction | E° , V |
|--|---------------|
| $\text{PbO}_2(\text{s}) + 4 \text{H}^+(\text{aq}) + 2e^- \rightarrow \text{Pb}^{2+}(\text{aq}) + 2 \text{H}_2\text{O}(\text{l})$ | 1.468 |
| $\text{Pb}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Pb}(\text{s})$ | -0.126 |
- (A) 0.671 V (B) 0.797 V
 (C) 1.342 V (D) 1.594 V
19. What is the average oxidation state of sulfur in calcium tetrathionate, CaS_4O_6 ?
- (A) -2.0 (B) +2.0 (C) +2.5 (D) +3.0

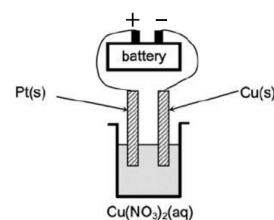
20. When the equation for the oxidation of $\text{Cu}(\text{s})$ by $\text{NO}_3^-(\text{aq})$ to give $\text{Cu}^{2+}(\text{aq})$ and $\text{NO}(\text{g})$ in acidic aqueous solution is balanced using smallest whole number coefficients, what is the coefficient for $\text{NO}_3^-(\text{aq})$?

(A) 1 (B) 2 (C) 3 (D) 6

21. Water is electrolyzed with a current of 10.0 A until 132 L of hydrogen gas (after drying, measured at STP) has been produced. How much time does this take?

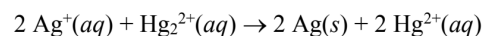
(A) 7.90 h (B) 15.8 h (C) 31.6 h (D) 63.2 h

22. A 1.0 M solution of $\text{Cu}(\text{NO}_3)_2$ is electrolyzed using the setup shown below. What reaction occurs at the cathode?



- (A) $\text{Pt}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Pt}(\text{s})$
 (B) $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Cu}(\text{s})$
 (C) $\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + 4e^- \rightarrow 4 \text{OH}^-(\text{aq})$
 (D) $2 \text{H}_2\text{O}(\text{l}) + 2e^- \rightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$

23. What is E° for the oxidation of mercury(I) ion by silver ion?



Half-reaction	E° , V
$\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag}(\text{s})$	+0.80
$2 \text{Hg}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Hg}_2^{2+}(\text{aq})$	+0.91

- (A) -0.11 V (B) 0.69 V
 (C) 1.71 V (D) 2.51 V

24. What is the standard reduction potential of $\text{PbF}_2(\text{s})$ at 298 K? The K_{sp} of PbF_2 at 298 K is 4.0×10^{-8} .

Half-reaction	E° , V
$\text{Pb}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Pb}(\text{s})$	-0.13
$\text{PbF}_2(\text{s}) + 2e^- \rightarrow \text{Pb}(\text{s}) + 2 \text{F}^-(\text{aq})$	???

- (A) -0.35 V (B) -0.22 V
 (C) -0.13 V (D) -0.04 V

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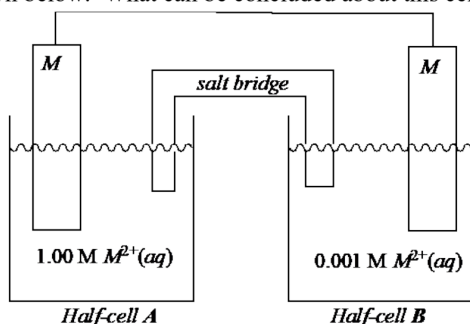
25. What is the average oxidation state of iron in riebeckite, $\text{Na}_2\text{Fe}_5\text{Si}_8\text{O}_{22}(\text{OH})_2$?

- (A) +2.0 (B) +2.4 (C) +3.0 (D) +5.6

26. An electrolytic cell is operated for 3000 s using a current of 1.50 A. From which 1.0 M solution will the smallest mass of metal be deposited?

- (A) CdCl_2 (B) InBr_3
(C) TlNO_3 (D) $\text{Pb}(\text{NO}_3)_2$

27. A galvanic cell using an unknown metal M is arranged as shown below. What can be concluded about this cell?



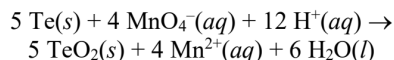
- (A) Half-cell A is the cathode.
(B) Half-cell B is the cathode.
(C) No current flows in the cell since the same metal is used in both half-cells.
(D) Which cell is the cathode depends on the standard reduction potential of $\text{M}^{2+}(\text{aq})$.

28. When manganese metal is placed in a solution containing silver(I) ion in acidic solution, metallic silver deposits on the manganese. Which half-reactions are taking place?

- I. $\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag}(\text{s})$
II. $\text{Mn}(\text{s}) + 4 \text{H}_2\text{O}(\text{l}) \rightarrow \text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 7e^-$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

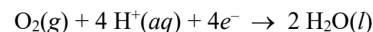
29. What is E° for the following redox reaction?



Half-reaction	E°, V
$\text{TeO}_2(\text{s}) + 4 \text{H}^+(\text{aq}) + 4e^- \rightarrow \text{Te}(\text{s}) + 2 \text{H}_2\text{O}(\text{l})$	+0.59
$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5e^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}$	+1.51

- (A) -3.09 V (B) 0.92 V
(C) 3.09 V (D) 5.19 V

30. The reduction of O_2 to H_2O in acidic solution has a standard reduction potential of +1.23 V. What is the effect on the half-cell potential at 25 °C when the pH of the solution is decreased by one unit?

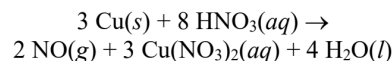


- (A) The half-cell potential decreases by 236 mV.
(B) The half-cell potential increases by 236 mV.
(C) The half-cell potential decreases by 59 mV.
(D) The half-cell potential increases by 59 mV.

31. Romanèchite has the formula $\text{Ba}_2\text{Mn}_5\text{O}_{10} \cdot 2\text{H}_2\text{O}$. What is the average oxidation state of manganese in romanèchite?

- (A) +2.0 (B) +3.2 (C) +3.6 (D) +4.0

32. Which element is being reduced in the following equation?



- (A) Cu (B) H (C) N (D) O

33. A galvanic cell is constructed under standard conditions using cobalt in cobalt(II) nitrate solution and indium in indium(III) nitrate solution. Which statements about this cell are correct?

Half-reaction	E°, V
$\text{Co}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Co}(\text{s})$	-0.28
$\text{In}^{3+}(\text{aq}) + 3e^- \rightarrow \text{In}(\text{s})$	-0.34

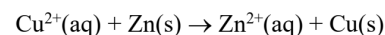
- I. The standard cell potential is 0.46 V.
II. Indium is the anode.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

34. Molten sodium chloride is electrolyzed for 10.0 minutes with a constant current of 10.0 A. What mass of sodium metal is produced?

- (A) 1.43 g (B) 2.20 g (C) 3.63 g (D) 4.41 g

35. The standard cell potential of a galvanic cell based on the reaction below is 1.10 V. Which concentrations would result in the largest measured voltage?



- (A) $[\text{Cu}^{2+}] = 1.0 \text{ M}$ and $[\text{Zn}^{2+}] = 1.0 \text{ M}$
(B) $[\text{Cu}^{2+}] = 3.0 \text{ M}$ and $[\text{Zn}^{2+}] = 1.0 \text{ M}$
(C) $[\text{Cu}^{2+}] = 1.0 \text{ M}$ and $[\text{Zn}^{2+}] = 3.0 \text{ M}$
(D) $[\text{Cu}^{2+}] = 3.0 \text{ M}$ and $[\text{Zn}^{2+}] = 3.0 \text{ M}$

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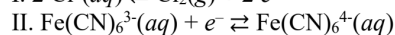
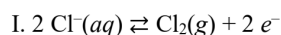
36. The standard reduction potential of oxygen under acidic conditions at 298 K is +1.23 V. What is the standard reduction potential for the four-electron reduction of $O_2(g)$ under basic conditions?

(A) 1.23 V (B) 1.02 V (C) 0.83 V (D) 0.40 V

37. What is the average oxidation state of tin in the mineral abhurite, $Sn_{21}Cl_{16}(OH)_{14}O_6$?

(A) +1.71 (B) +2.00 (C) +2.76 (D) +3.43

38. If used in an electrochemical cell, which of the following half-reactions would require an inert electrode?



(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

39. Chromium is electroplated industrially by the electrolysis of solutions of $K_2Cr_2O_7$. How much time would be required to deposit 1.00 kg of Cr using a current of 200.0 A?

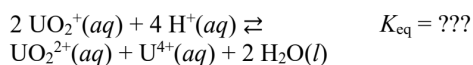
(A) 2.58 h (B) 7.74 h (C) 15.5 h (D) 31.0 h

40. What is the standard reduction potential of $Cr^{3+}(aq)$ to form metallic chromium?

Half-reaction	E° , V
$Cr^{3+}(aq) + e^- \rightarrow Cr^{2+}(aq)$	-0.41
$Cr^{2+}(aq) + 2e^- \rightarrow Cr(s)$	-0.91
$Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$	???

(A) -0.74 V (B) -1.32 V
(C) -1.73 V (D) -2.23 V

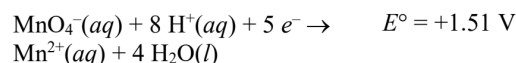
41. What is the equilibrium constant at 298 K for the disproportionation of uranium(V) in acidic solution as shown below?



Half-reaction	E° , V
$UO_2^{2+}(aq) + e^- \rightarrow UO_2^+(aq)$	+0.16
$UO_2^+(aq) + 4 H^+(aq) + e^- \rightarrow U^{4+}(aq) + 2 H_2O(l)$	+0.27

(A) 1.9×10^{-4} (B) 73
(C) 5300 (D) 1.9×10^7

42. The standard reduction potential E° for the reduction of permanganate in acidic solution is +1.51 V. What is the reduction potential for this half-reaction at pH = 5.00?

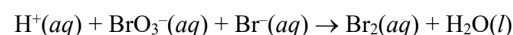


(A) +1.50 V (B) +1.42 V
(C) +1.04 V (D) -0.85 V

43. What is the average oxidation state of tungsten in sodium phosphotungstate, $Na_3PW_{12}O_{40}$?

(A) +6.00 (B) +6.17
(C) +6.42 (D) +6.67

44. Bromate ion and bromide ion react to form bromine in acidic solution. When the reaction is balanced, which statement about $H^+(aq)$ is correct?



(A) Its coefficient is twice the coefficient of $Br^-(aq)$.
(B) Its coefficient is twice the coefficient of $Br_2(aq)$.
(C) Its coefficient is twice the coefficient of $BrO_3^-(aq)$.
(D) Its coefficient is the sum of the coefficients of $BrO_3^-(aq)$ and $H_2O(l)$.

45. Which reactions are spontaneous under standard conditions?

Half-Reaction	E° , V
$Cu^+(aq) + e^- \rightarrow Cu(s)$	+0.52
$Co^{2+}(aq) + 2 e^- \rightarrow Co(s)$	-0.28
$In^{3+}(aq) + 3 e^- \rightarrow In(s)$	-0.34

I. $2 Cu^+(aq) + Co(s) \rightarrow Co^{2+}(aq) + 2 Cu(s)$
II. $3 Co^{2+}(aq) + 2 In(s) \rightarrow 2 In^{3+}(aq) + 3 Co(s)$

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

46. An electrolytic cell consists of two copper electrodes immersed in a solution of copper(II) sulfate. What is the result of passing 0.35 A of current through this cell for 1300 s?

(A) The mass of the anode increases by 0.15 g.
(B) The mass of the anode increases by 0.30 g.
(C) The mass of the anode decreases by 0.15 g.
(D) The mass of the anode decreases by 0.30 g.

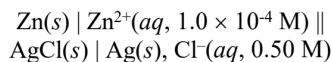
Oxidation - Reduction

47. What is the change in standard free energy at 298 K for the conversion of ozone to molecular oxygen as shown in the equation below?



Half-Reaction	E°, V
$\text{O}_3(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 e^- \rightarrow \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	+2.08
$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 e^- \rightarrow 2 \text{H}_2\text{O}(\text{l})$	+1.23

- (A) -164 kJ mol^{-1} (B) -328 kJ mol^{-1}
 (C) -401 kJ mol^{-1} (D) -492 kJ mol^{-1}
48. What is the cell potential of the following galvanic cell under the stated conditions at 298 K?



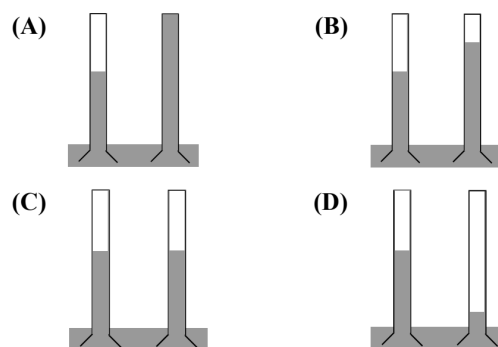
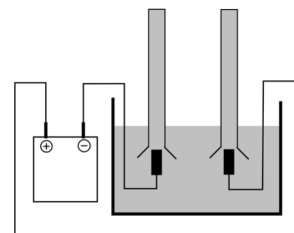
Half-Reaction	E°, V
$\text{Zn}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Zn}(\text{s})$	-0.762
$\text{AgCl}(\text{s}) + e^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.222

- (A) +0.884 V (B) +0.984 V
 (C) +1.084 V (D) +1.120 V
49. When the chemical equation is balanced, what is the ratio of the coefficient of $\text{Ag}(\text{s})$ to the coefficient of $\text{H}_2\text{O}(\text{l})$?
- $$__ \text{Ag}(\text{s}) + __ \text{NO}_3^-(\text{aq}) + __ \text{H}^+(\text{aq}) \rightarrow __ \text{Ag}^+(\text{aq}) + __ \text{NO}(\text{g}) + __ \text{H}_2\text{O}(\text{l})$$
- (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 3 : 2
50. A current of 0.44 A is passed through a solution of a ruthenium nitrate salt, causing reduction of the metal ion to the metal. After 25.0 minutes, 0.345 g of $\text{Ru}(\text{s})$ has been deposited. What is the oxidation state of ruthenium in the nitrate salt?
- (A) +2 (B) +3 (C) +4 (D) +6
51. What is the K_{sp} of $\text{Ca}(\text{OH})_2$ at 298 K?

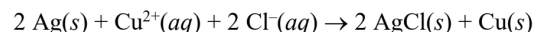
Half-Reaction	E°, V
$\text{Ca}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Ca}(\text{s})$	-2.87 V
$\text{Ca}(\text{OH})_2(\text{s}) + 2 e^- \rightarrow \text{Ca}(\text{s}) + 2 \text{OH}^-(\text{aq})$	-3.02 V

- (A) 1.6×10^{-6} (B) 3.1×10^{-6}
 (C) 8.4×10^{-6} (D) 2.9×10^{-3}

52. Electrolysis of water containing sulfuric acid as an electrolyte is carried out as shown and the gases produced at the electrodes collected in two identical tubes, both initially filled with the sulfuric acid solution. After electrolysis has been carried out for a certain time, which picture best represents the appearance of the tubes?

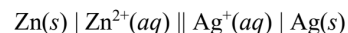


53. What is ΔE° for this reaction?

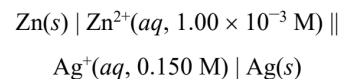


Half-Reaction	E°, V
$\text{AgCl}(\text{s}) + e^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	0.222 V
$\text{Cu}^{2+}(\text{aq}) + 2 e^- \rightarrow \text{Cu}(\text{s})$	0.337 V

- (A) -0.115 V (B) -0.107 V
 (C) 0.107 V (D) 0.115 V
54. The standard cell potential of the following galvanic cell is 1.562 V at 298 K.



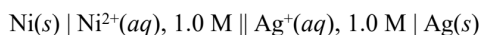
What is the cell potential of the following galvanic cell at 298 K?



- (A) 1.602 V (B) 1.626 V
 (C) 1.642 V (D) 1.691 V

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55. The mineral crocidolite has the formula $\text{Na}_2\text{Fe}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$. How many of the iron ions per formula unit are in the +2 oxidation state and how many are in the +3 oxidation state?
- (A) All five are +2
(B) Three are +2, two are +3
(C) Two are +2, three are +3
(D) One is +2, four are +3
56. What products are formed in the electrolysis of 1.0 M aqueous HBr?
- (A) H_2 at the cathode, Br_2 at the anode
(B) O_2 at the cathode, H_2 at the anode
(C) OH^- at the cathode, HOBr at the anode
(D) Br_3^- at the cathode, HBrO_4 at the anode
57. A 1.00 g sample of a hydrogen peroxide (H_2O_2) solution is placed in an Erlenmeyer flask and diluted with 20 mL of 1 M aqueous sulfuric acid. To this solution is added 0.0200 M KMnO_4 solution via a buret, until a pale purple color persists. This requires 22.50 mL of KMnO_4 solution. What is the percent by mass of hydrogen peroxide in the original solution?
- (A) 0.613% (B) 1.53% (C) 3.83% (D) 7.65%
58. What is the standard cell potential of the following electrochemical cell?



Half-Reaction	E°, V
$\text{Ni}^{2+}(aq) + 2e^- \rightarrow \text{Ni}(s)$	-0.23
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80

- (A) -1.83 V (B) 0.57 V (C) 1.03 V (D) 1.83 V

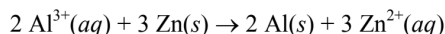
59. Copper electrodes are placed into two aqueous solutions of copper(II) sulfate at 25 °C. One compartment contains a 1.0 M solution while the other compartment contains a 0.10 M solution. The two compartments are connected with a salt bridge and the electrodes are connected by a wire passing through a voltmeter. In what direction do the electrons flow through the wire, and what is the cell potential read on the voltmeter?

Direction of electron flow	Cell potential
(A) From the electrode in the 1.0 M solution to the electrode in the 0.10 M solution	30 mV
(B) From the electrode in the 1.0 M solution to the electrode in the 0.10 M solution	59 mV
(C) From the electrode in the 0.10 M solution to the electrode in the 1.0 M solution	30 mV
(D) From the electrode in the 0.10 M solution to the electrode in the 1.0 M solution	59 mV

60. A 1.00 g sample of a silver-containing ore is dissolved in dilute nitric acid. The solution is neutralized and then selectively electrolyzed to deposit the silver metal, requiring 670 s of 0.10 A current. What is the mass percentage of silver in the ore?
- (A) 2.5% (B) 3.0% (C) 3.7% (D) 7.5%
61. Which is formed at the cathode during the electrolysis of aqueous AgF ?
- (A) $\text{Ag}(s)$ (B) $\text{H}_2(g)$ (C) $\text{O}_2(g)$ (D) $\text{F}_2(g)$
62. In which of the following substances is chlorine in the lowest oxidation state?
- (A) Cl_2 (B) KCl (C) KClO (D) KClO_4
63. When the following reaction is balanced, what is the ratio of coefficients of $\text{H}^+(aq)$ to $\text{NO}(g)$?
- $$\text{Cu}(s) + \text{H}^+(aq) + \text{NO}_3^-(aq) \rightarrow \text{NO}(g) + \text{H}_2\text{O}(l) + \text{Cu}^{2+}(aq)$$
- (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 4 : 1

Oxidation - Reduction

64. What is E° for the following reaction?



Half-Reaction	E° , V
$\text{Al}^{3+}(aq) + 3 e^- \rightarrow \text{Al}(s)$	-1.66
$\text{Zn}^{2+}(aq) + 2 e^- \rightarrow \text{Zn}(s)$	-0.76

- (A) -1.04 V (B) -0.90 V
(C) 0.90 V (D) 1.04 V

65. Electrolysis of 1.00 g of a copper(II) salt required passage of 0.100 A of current for 123 min for complete deposition of all the copper metal. What is the copper salt?

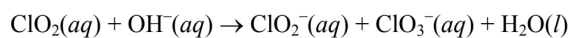
- (A) CuCl_2 , $M = 134.5$ (B) CuBr_2 , $M = 223.4$
(C) $\text{Cu}(\text{NO}_3)_2$, $M = 187.6$ (D) $\text{Cu}(\text{ClO}_4)_2$, $M = 262.5$

66. What is the K_{sp} of Hg_2Cl_2 at 298 K?

Half-Reaction	E° , V (at 298 K)
$\text{Hg}_2^{2+} + 2e^- \rightarrow 2 \text{Hg}(l)$	+0.80
$\text{Hg}_2\text{Cl}_2(s) + 2e^- \rightarrow 2 \text{Hg}(l) + 2 \text{Cl}^-(aq)$	+0.31

- (A) 2.6×10^{-17} (B) 3.3×10^{-11}
(C) 5.1×10^{-9} (D) 5.7×10^{-6}

67. When the equation



is balanced, what is the ratio of the coefficient of ClO_2 to that of ClO_3^- ?

- (A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 3 : 2

68. When an aqueous solution of KI is electrolyzed, what forms at the anode?

- (A) O_2 (B) I_2 (C) K (D) H_2O

69. A current of 0.15 A is passed through an aqueous solution of K_2PtCl_4 . How long will it take to deposit 1.00 g Pt(s) ($M = 195.1$)?

- (A) 1600 s (B) 3300 s (C) 6600 s (D) 13000 s

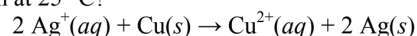
70. In the galvanic cell



which of the following changes will increase the cell potential?

- I. Dilution of the Al^{3+} solution to 0.001 M
II. Dilution of the Cu^{2+} solution to 0.001 M
III. Increasing the surface area of the $\text{Al}(s)$ electrode
(A) I only (B) II only
(C) III only (D) I and III only

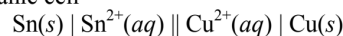
71. What is the equilibrium constant for the following reaction at 25 °C?



Half-Reaction	E° , V
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	+0.80
$\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$	+0.34

- (A) 6.0×10^7 (B) 3.6×10^{15}
(C) 3.6×10^{38} (D) 4.2×10^{42}

72. In the galvanic cell



the standard potential is 0.48 V. Starting with standard concentrations, what are the concentrations of Sn^{2+} and Cu^{2+} when the cell has discharged to a potential of 0.45 V?

- (A) $[\text{Sn}^{2+}] = 0.47 \text{ M}$, $[\text{Cu}^{2+}] = 1.53 \text{ M}$
(B) $[\text{Sn}^{2+}] = [\text{Cu}^{2+}] = 1.00 \text{ M}$
(C) $[\text{Sn}^{2+}] = 1.53 \text{ M}$, $[\text{Cu}^{2+}] = 0.47 \text{ M}$
(D) $[\text{Sn}^{2+}] = 1.82 \text{ M}$, $[\text{Cu}^{2+}] = 0.18 \text{ M}$